

METHODOLOGY

Software Requirements Specification

Product Scope

The objective of this study is to develop a web-based Document Tracking System for the Ethical Review Board of Holy Name University. The project is focused on providing solutions for the following processes: Document Submissions, Submission Status Tracking, Reviewer Management, and Submission Issues Management. This project also involved the different stakeholders for the Ethical Review Board which includes the ERB Head, ERB staff, Reviewers, and Research advisers.

The project was developed using a local environment development server. After the development, testing was done to ensure that the system meets the user's expectations. To determine the respondents for the testing, the researchers used non-probability sampling methods, specifically: purposive sampling and convenience sampling. In purposive sampling, respondents are selected since they already have the characteristics the researchers need for a sample. [13] The identified respondents are the ERB Head, the ERB Reviewers, Staff, and Research Advisers, which possess the necessary knowledge and can provide insights relevant to the system. In convenience sampling, respondents were selected because they were the easiest for the researchers to access. [14]

For the advisers, the researchers chose to test the faculty members of his own department. For the reviewers, the researchers sent an invitation to test the system through email prior to the testing. The reviewers that responded through the email were prioritized. For those who did not respond through the email, the researchers visited them in their respective offices and were asked if they were willing to participate in the test and schedule a time suitable for them.

Usability test was performed for the testing. The survey tool developed by usability.org was used to measure the usability of the system, which is composed of 10 statements that let the user rate each from Strongly Disagree to Strongly Agree. Refer to Appendix A for the details of the questionnaire. The researchers used the System Usability Scale, a reliable tool for measuring the usability of

various products and services including hardware, software, mobile devices, websites, and applications. [15].

Both the ERB-Head and Staff willingly participated in the test. For the advisers, there were a total of seven (7) testers. For the reviewers, ten (10) individuals participated in the test, with a total of nineteen (19) respondents.

The software testing was done personally. The researchers used a functionality test to ensure that the system functionalities work for a specific user. During testing, the researchers gathered feedback from the responses for improvements to be made to the system.

The nineteen (19) responses were analyzed using descriptive statistics. The graphs generated by Google Forms were used for the summary.

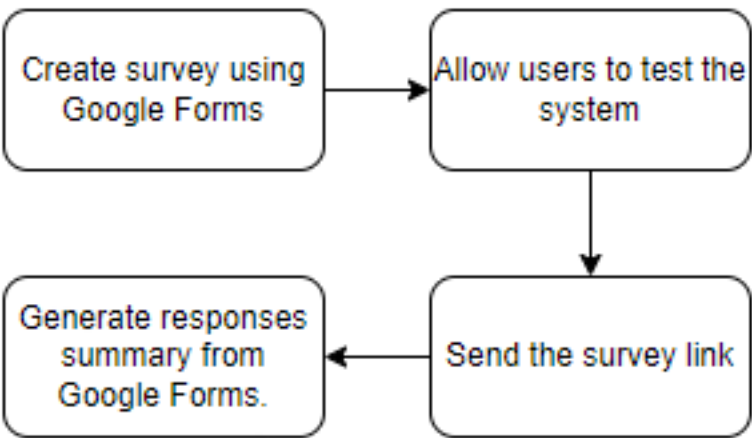


Figure 6. Data Collection and Analysis Process

The use of google forms in collecting survey responses was done to minimize the cost of using pen and paper and streamline the analysis of the collected data. The results of the Usability Test were used to develop the application further.

Functional Requirements

The software was divided into five main functionalities: Manage Accounts, Manage Submissions, Manage Reviewers, Manage Submission Issues, and Generate Reports.

Manage accounts involves the management of user accounts in the software. This includes registering / adding of new accounts, updating account information, and disabling of accounts.

Manage Submissions involves the management of Documents submitted to the ERB. This includes adding, updating, and deleting new submission records. This also includes the adding of documents for a specific submission.

Manage Reviewers involves the assigning of reviewers for a submission. This allows users to add and remove a reviewer to review a submission.

Manage Submission issues allows the management of submission issues. This allows the users to add, update, and delete submission issues. This also allows users to respond and resolve issues.

Generate Reports provide users with generic information of submissions. This also allows the admin to view detailed submission reports.

Non-Functional Requirements

The system provides various non-functional capabilities for users to have a great user experience. One important functionality is Portability and Compatibility. The system is tested and can be used in multiple browsers. This ensures that whatever browser the user prefers, they can still be able to use the system. It also provides a responsive design that adjusts the user interface depending on the screen size of the user.

Another non-functional capability the system provides is Data Integrity. This ensures that data entered in the various forms in the system are validated to avoid erroneous data in the database. New users, specifically advisers, are required to verify their email to ensure that the users stored in the database are also validated.

Usability is also another non-functional capability of the system. The system follows conventional User Experience patterns to ensure that users can use the system with little to no help from a technical individual.

Security is an important aspect that the system also provides. To ensure that the system is secured, passwords are encrypted so that it cannot be accessed directly in case of a database breach. The system also uses Laravel's

authentication feature which ensures that users that login to the system are verified and has access rights to the resource that it is accessing into.

Process Modeling

Business Process Model and Notation (BPMN) is utilized to visually represent the step-by-step business operations of the Holy Name University Ethical Review Board (HNU-ERB). For an Information Systems perspective, analyzing the workflow is crucial to identifying operational bottlenecks. This approach provides a clear graphical view of the end-to-end flow of the document tracking and review process, highlighting the efficiency gains introduced by the proposed automated system compared to the manual counterpart.

Current Process (Manual Workflow)

Currently, the HNU-ERB follows a fragmented approach to document routing. The researchers must submit a physical hard copy to the ERB staff, while the research adviser simultaneously emails a soft copy using a designated tracking code. Upon receipt, the ERB Head manually assigns the document to a reviewer. The recording and tracking of these submissions heavily rely on manual data entry into Google Sheets and organizing emails via Gmail "Tags". Because this process is disjointed, files and emails are occasionally misplaced due to typographical errors in tags, and researchers are forced to physically visit the office to inquire about their submission status.

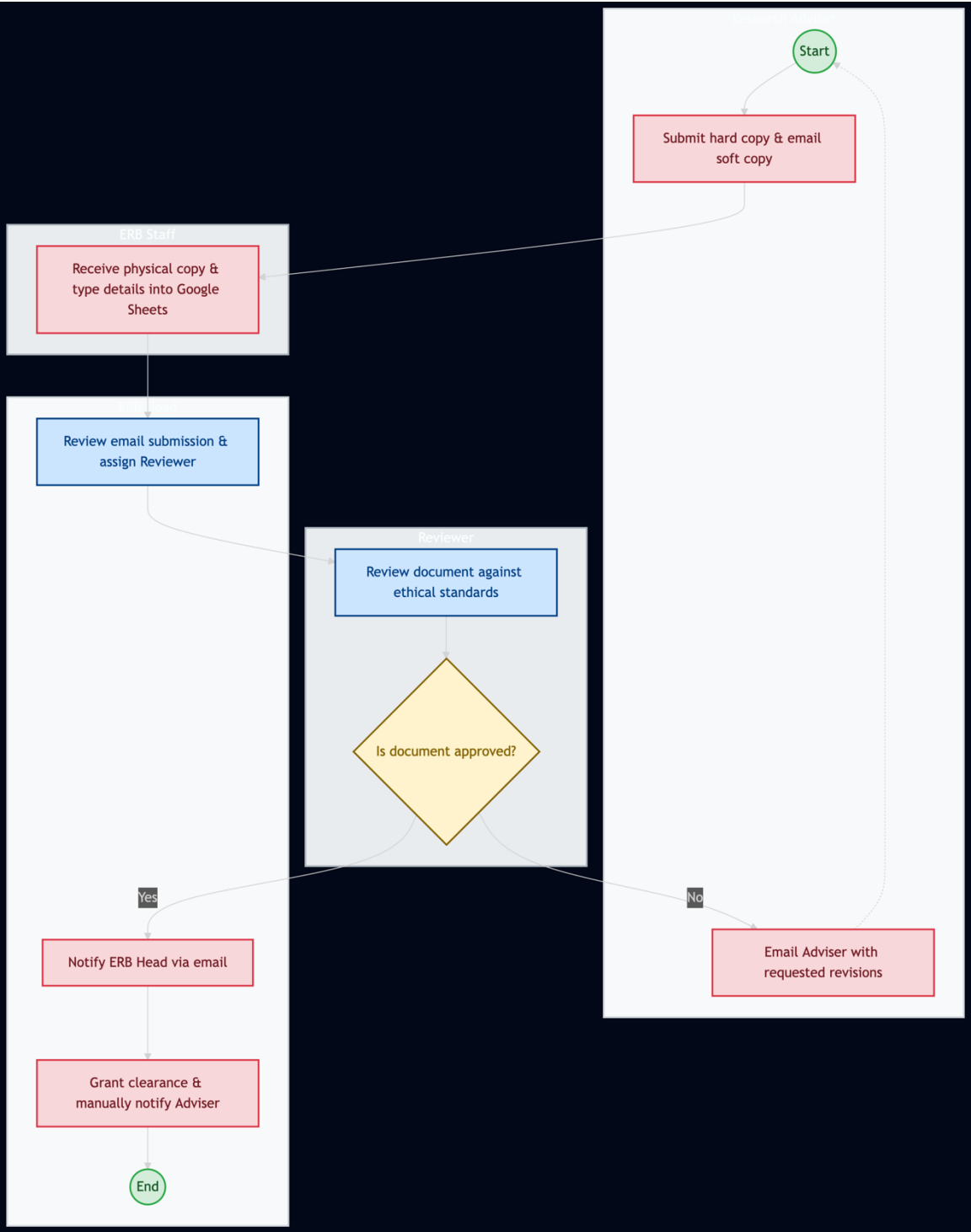


Figure 7. BPMN Diagram of Current (Manual) Process

Proposed Process (Automated Workflow)

The proposed TrackERB system transitions the workflow into a centralized Electronic Document Management System (EDMS) that actively manages the routing. Research advisers submit documents directly through the secure web platform, eliminating email dependency and physical submissions. The system automatically tags statuses and facilitates reviewer assignments. Reviewers utilize a dedicated "Issues Management" module to digitally tag specific errors directly in the system for the author to revise.

Furthermore, automated SMS and email notifications proactively inform users of required actions, replacing manual follow-ups.

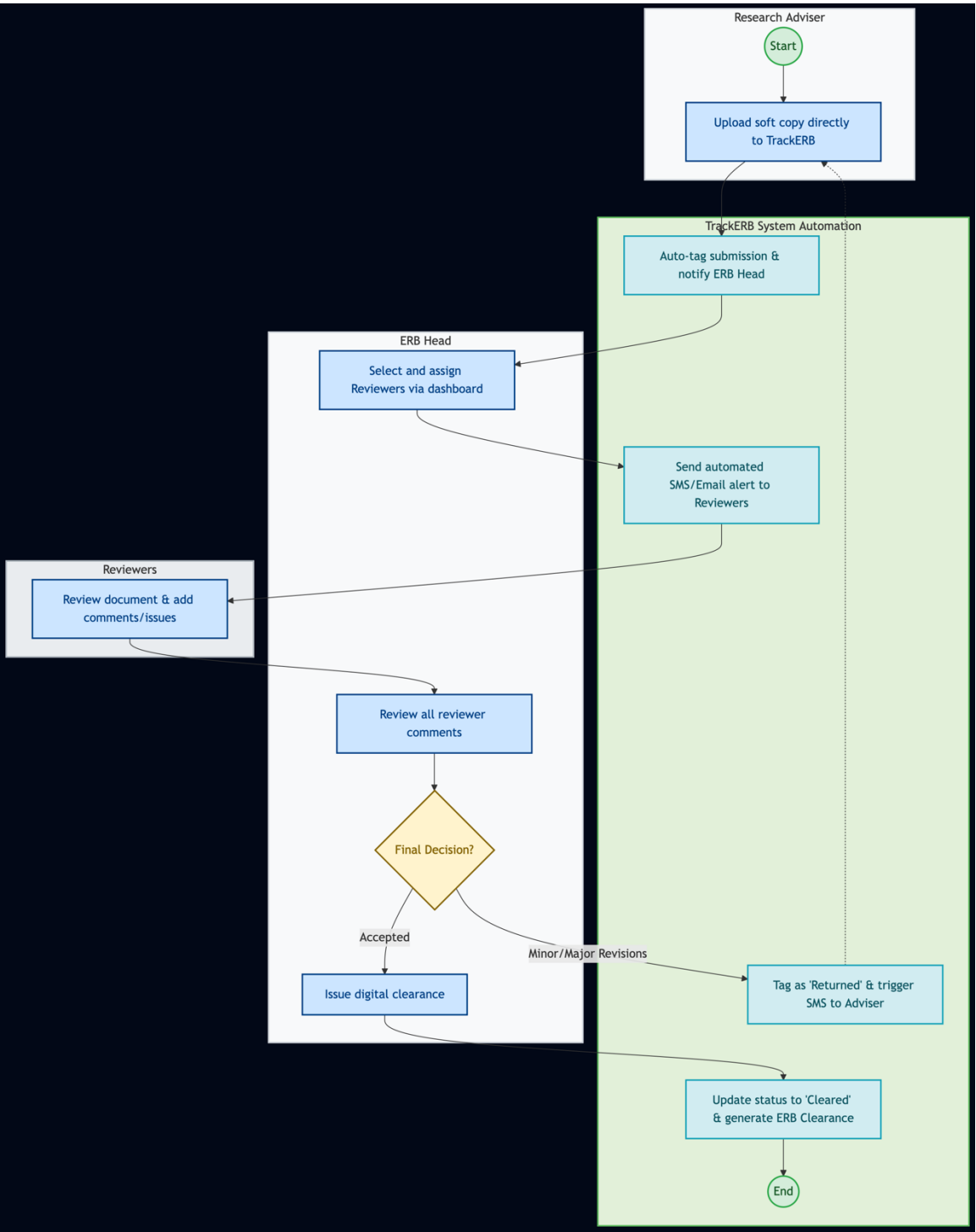


Figure 8. BPMN Diagram of Proposed (Automated) Process

Project Design

This section shows the various architecture and technologies used in the development of the system.

Laravel and the MVC Architecture

The system was developed using Laravel 10, a PHP framework that was developed to make the development process standardized, automatically processing some non-business logic relationship and allowing programmers to focus on implementing business logic.[16]

Laravel uses the Model-View-Controller architecture. The MVC architecture is a three-way division of an application whereby objects of different classes take over the operations related to the application domain (model), the display of the application's state (view), and the user interaction with the model and the view (controller). [17]

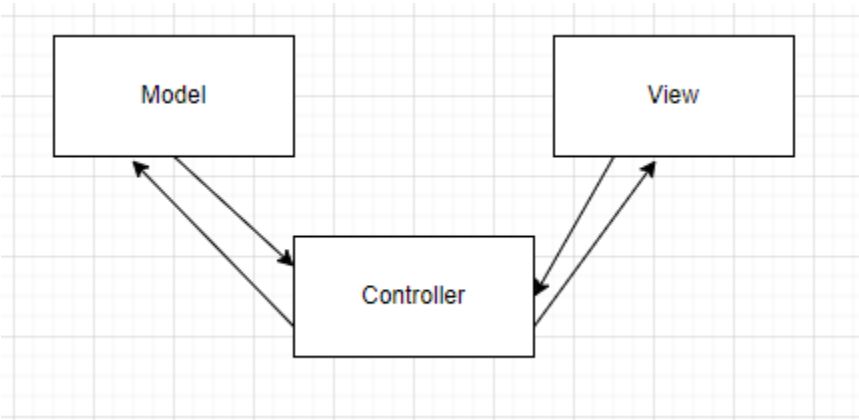


Figure 9. MVC Architecture

Frameworks and APIs

The system was created using various Frameworks and APIs. Tailwind CSS, a CSS framework that uses utility classes to generate styles and write them on a static CSS file, was used to develop the front end along with Blade, Laravel's templating engine. Chart.JS, a charting library, was used for the generation of charts and reports. For the back end, the system uses Laravel's built-in functionalities. Eloquent, Laravel's built-in object-relational mapper, was used to interact with the database. Dompdf, an HTML-to-PDF converter, was used to generate pdf files. Vonage, a flexible communications platform, was used to send SMS notifications to users using its Communications API. The system uses MySQL for the database. Laragon was used for the development environment for the system, which works well with Laravel and MySQL.

Use Case Diagram

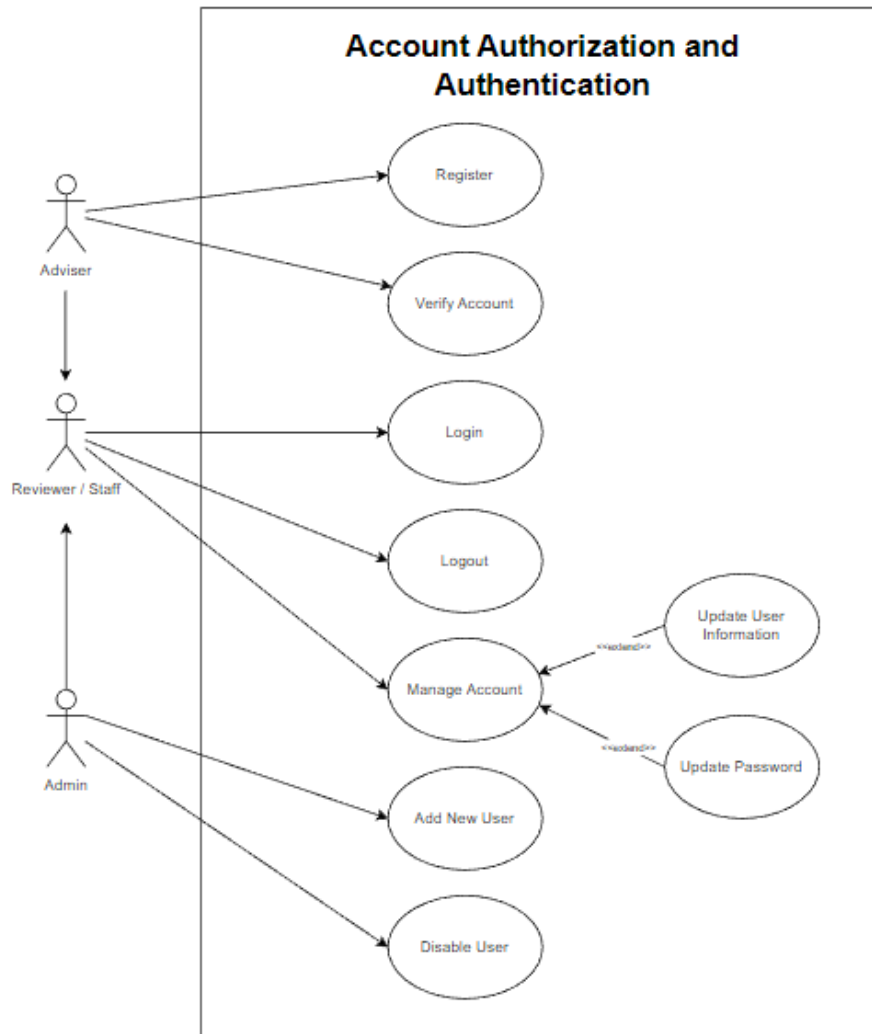


Figure 10. Account Authorization and Authentication

Account authorization and authentication involve all the actors. Before using the system, the adviser needs to register first. After registering, the system will send a verification email. After clicking the link, the user is automatically logged in to the system. Only the admin can register accounts for Reviewers and Staff. All actors can manage their own accounts. They can update their account information as well as update their passwords. The admin can also disable an account.

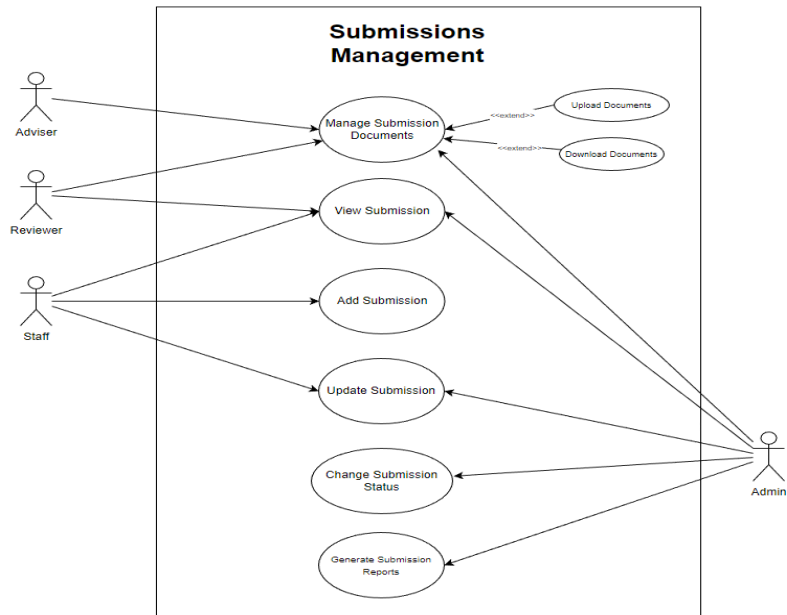


Figure 11. Submissions Management

Management of Submissions involves all actors. Initially, the adviser must submit hard copies of the research documents to the staff. The staff then enters the necessary information into the system. The staff can also update the submission information. After adding the submission, the adviser must submit a soft copy of the documents in the system. Upon uploading, admin and staff can download and view the documents. Reviewers can only download submission documents if they are assigned to them. The admin can update submission status and view reports generated from the submission data stored in the database.

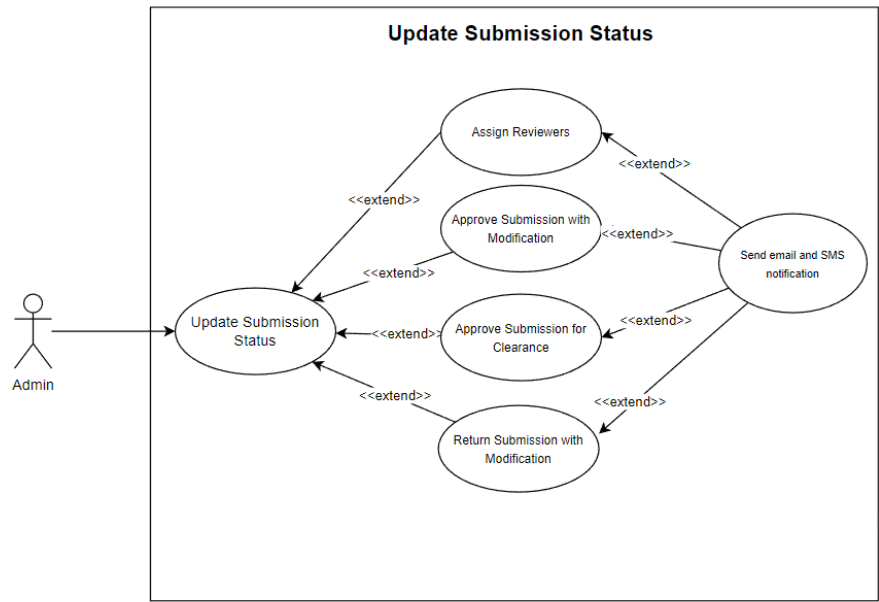


Figure 12. Update Submission Status (Admin)

Updating of submission status only involves the admin. The admin can assign reviewers for a specific submission. The admin can then decide whether to approve the submission, approve the submission but with pending modification, where the adviser can already print a clearance. However, they still need to modify the submission or return the submission with modifications where the adviser needs to revise their submission and resubmit before they can print the ERB clearance.

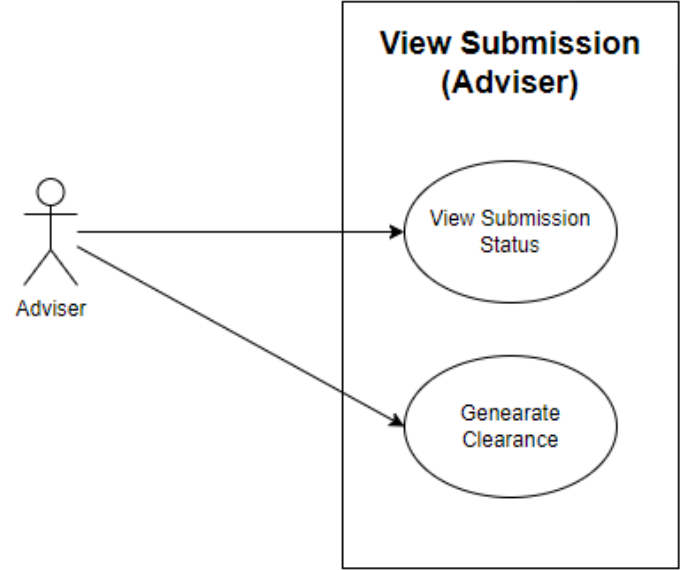


Figure 13. View Submission (Adviser)

In this use case, the adviser can track the status of their submissions. They can see if their status is pending, under review, approved, approved with modifications, or returned with modifications. If their submission is approved, they are able to print the ERB clearance.

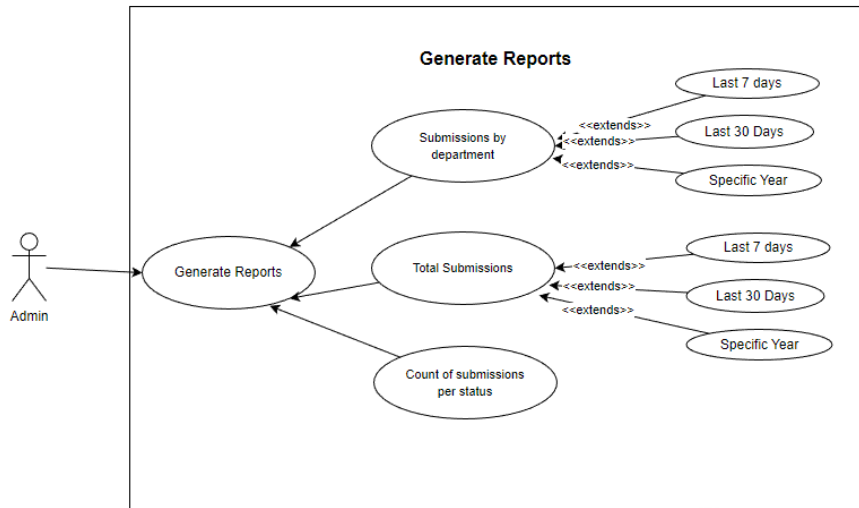


Figure 14. Generate Reports

This use case involves the admin. The admin is provided with a dashboard where the actor can see reports for the system. The reports include a submission by department and total submissions, which can be filtered by last 7 days, last 30 days, and specific year where they can choose a specific year. The actor is also provided with the count of submissions on each status.

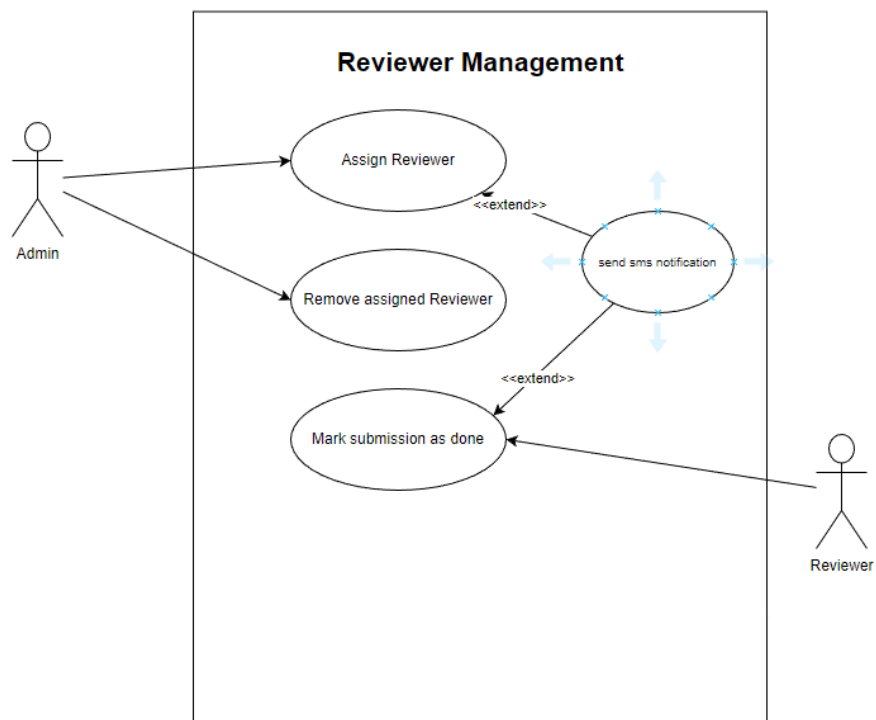


Figure 15. Reviewer Management

This use case involves the admin and reviewer. The admin can assign reviewers to a specific submission. Upon assigning, the reviewers will receive an SMS and Email notification to inform them that they are assigned a new

submission to review. The admin can also remove a reviewer, but only if the reviewer has not yet added an issue or change the review status to 'done.' After all the reviewers have completed their review, the system will send an SMS notification to the admin to notify them that the reviewers are already done reviewing the submission.

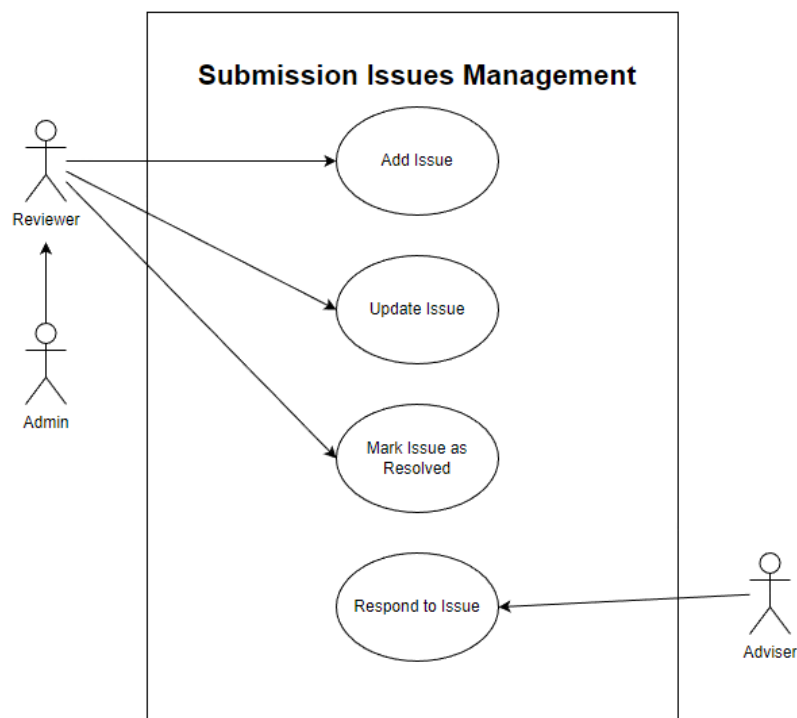


Figure 16. Submission Issues Management

This use case involves the reviewer, admin, and adviser. Mainly, the reviewer manages the issues for the submission. They can add and update issues. The submission must be tagged as 'approved with modifications' or 'returned with modifications' before the adviser can respond to the issue. After responding, the reviewer can then mark the issue as resolved. Both reviewer and admin have the same capabilities; the only difference is that the reviewer can only manage their own issues while the admin can manage all the issues.

Entity Relationship Diagram

The Entity Relationship Diagram shows the relationship, cardinalities, associations, and data types of the entities used by the system.

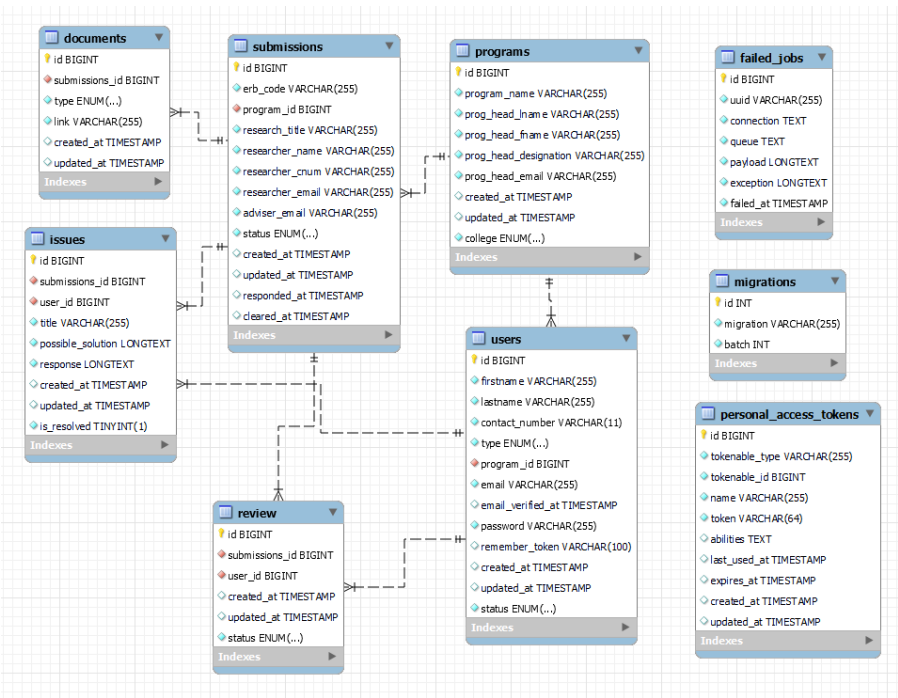


Figure 17. TrackERB Entity Relationship Diagram

Activity Diagram

The system currently has four (4) users namely ERB Head, Staff, Reviewer, and Adviser. The following figures will show the activity diagram that shows how these users interact with the system during a specific functionality or scenario.

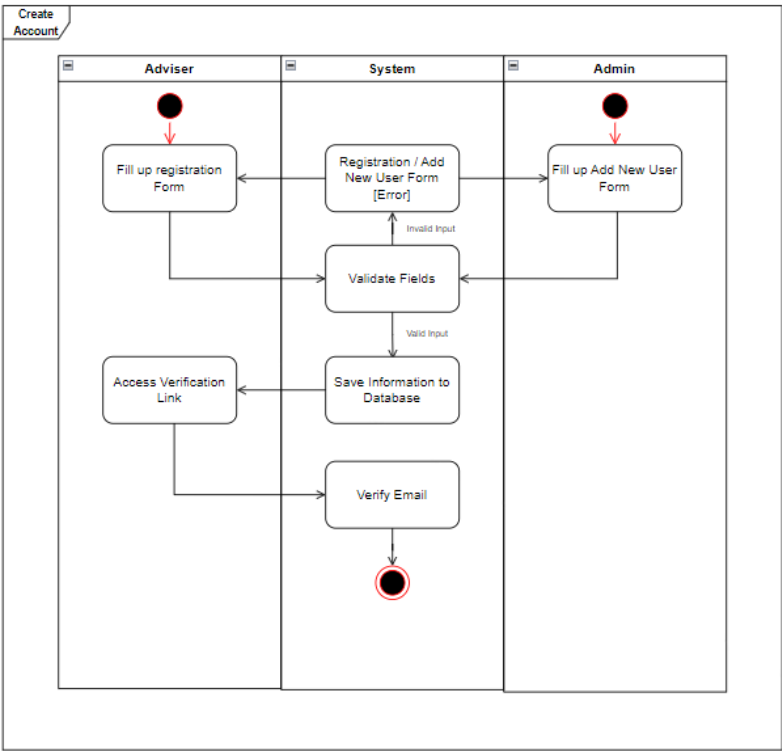


Figure 18. Register / Create New User

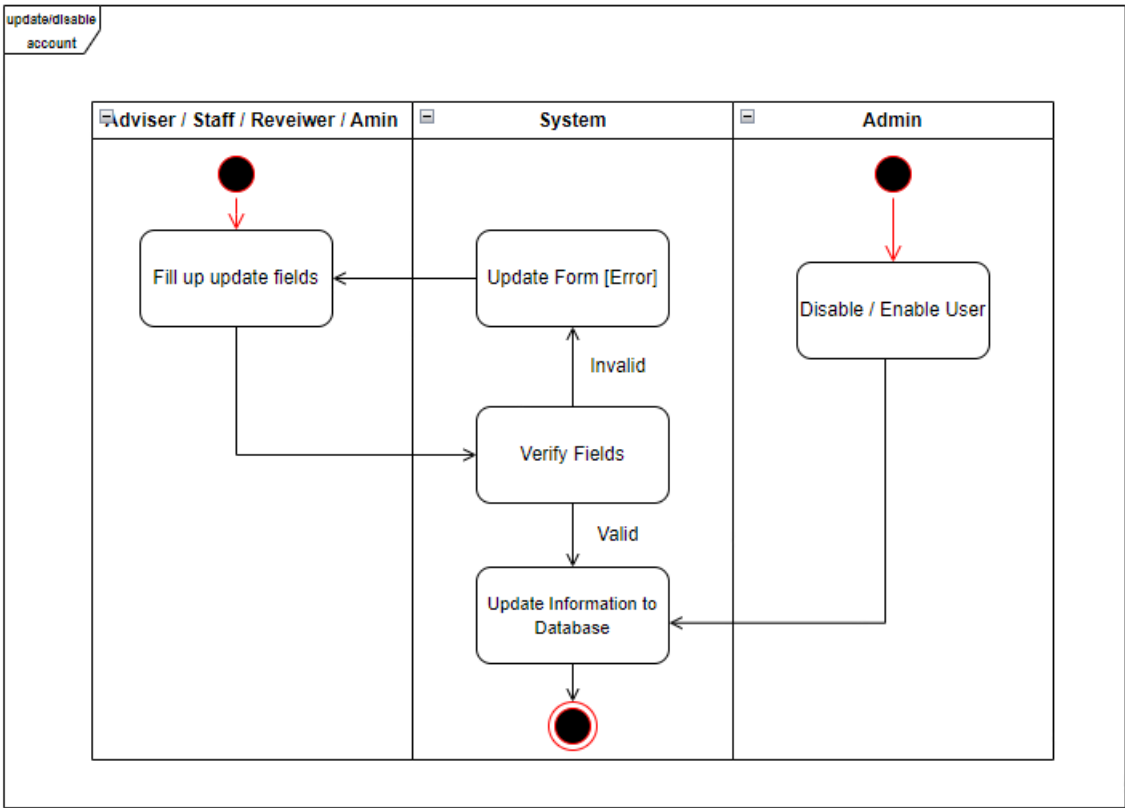


Figure 19. Update / Disable User

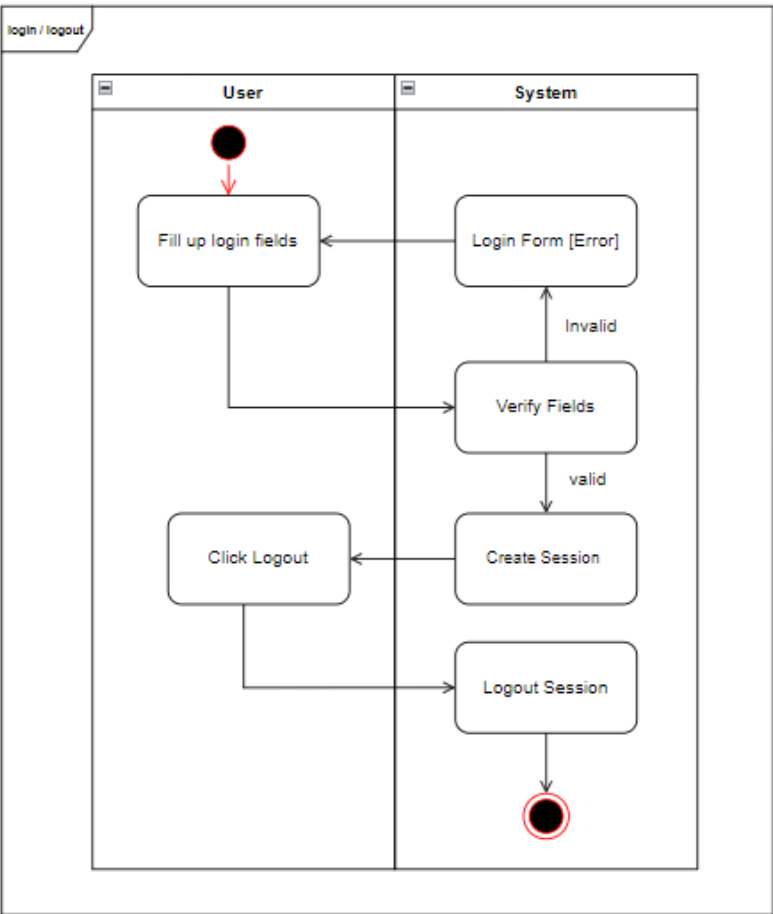


Figure 20. Login and Logout

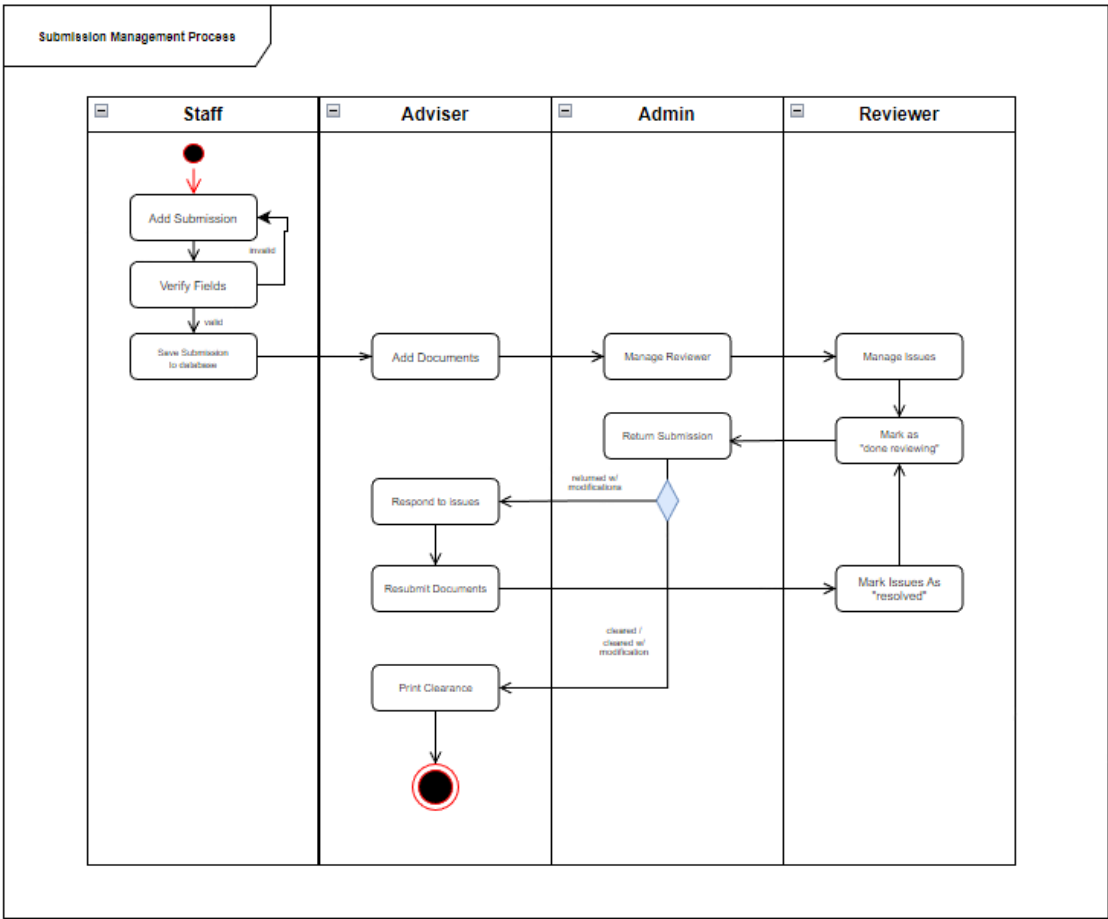


Figure 21. Submission Management Process

Application Prototype

This section shows the application’s prototype. This served as a reference for the final design of the product.

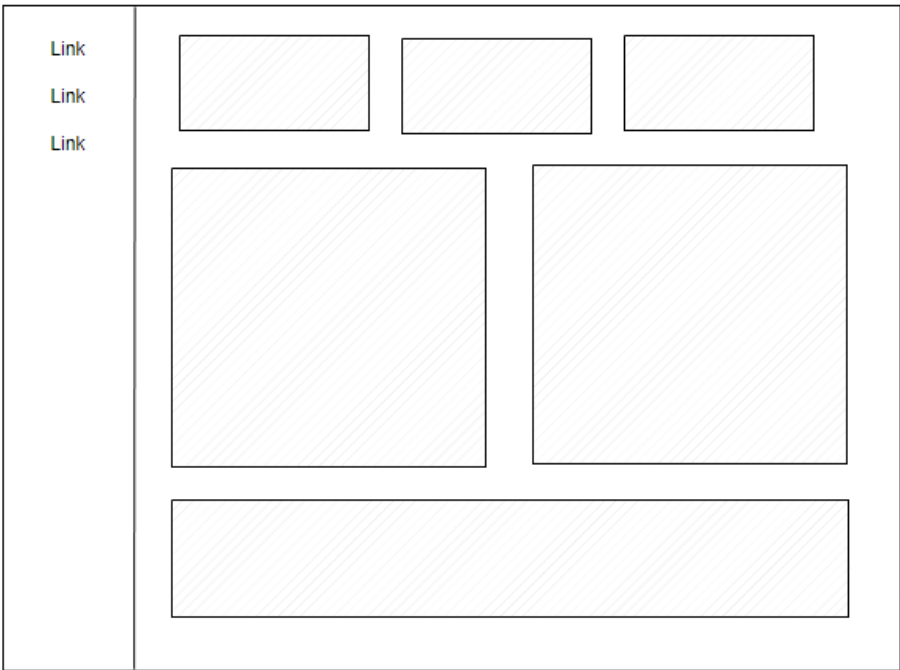


Figure 22. User Dashboard Page

Link Link Link	Name, Name	
	Repository	Collaborators
	3rd_library	+6
	ally-dashboard	
	aac-sac	
	3rd_library	+6

Figure 23. Data List Table Page

Link	Name, Name 										
Link	ERB-CODE										
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Figure 24. Submission Information Page

NavNavNav

Status 1

Status Description. This provides a status description.

|

Status 2

Status Description. This provides a status description.

|

Status 3

Status Description. This provides a status description.

|

Status 4

Status Description. This provides a status description.

ERB CODE

Name

Description

Email

Number

Figure 25. Submission Status Tracking Page

Name

Title

Email

College

Department

Conatct Person

ERB-CODE

Submit

Figure 26. Form Design